

2018 HKDSE Physics

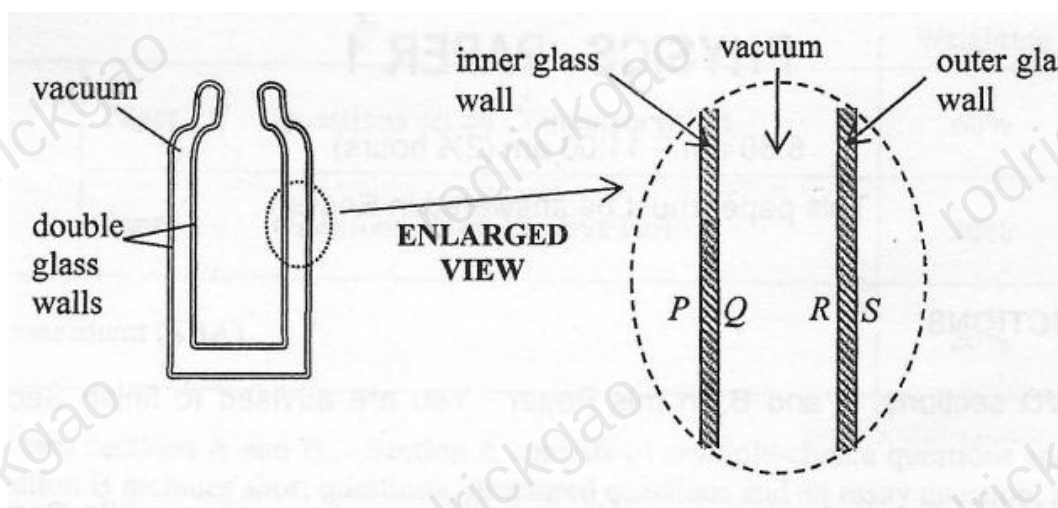
Paper 1A — Multiple-choice Questions

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Section A. There are 33 questions. Questions marked with * involve knowledge of the extension component. Question 1 was deleted/withdrawn in the official paper.

Question 1 (Withdrawn)

The figure shows a vacuum flask with double glass walls for keeping liquids cold. P, Q and R, S are the glass surfaces of the inner and outer glass walls respectively. Which two surfaces are usually coated with silver?



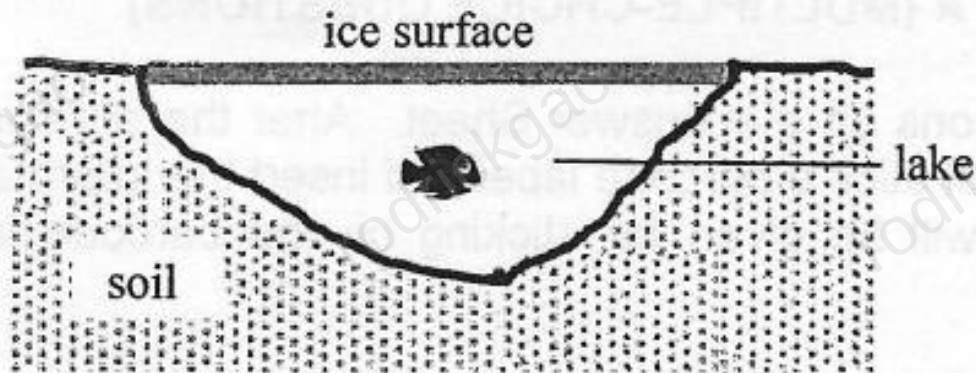
- A P and R
- B Q and R
- C P and S
- D R and S

Official answer: Deleted

This item was officially withdrawn, so no official answer is assigned and no replacement choice is inferred.

Question 2

In some countries, the outdoor temperature can drop below 0°C in winter such that the lake surface becomes a thick layer of ice. However the water beneath the ice surface does not easily turn into ice and most aquatic life can survive during winter time. Which statement below best explains this phenomenon?



- A The layer of ice provides good heat insulation.
- B The freezing point of water below the ice surface is much lower than 0°C .
- C There is thermal energy transferred from the soil to the water in the lake.
- D Ice releases latent heat when it melts.

Official answer: A

Ice has low thermal conductivity, so the surface layer reduces heat transfer from the water below to the cold air. The underlying water therefore cools and freezes much more slowly. Answer: A.

Question 3

Milk at $5\text{ }^{\circ}\text{C}$ is added to a cup of tea at $25\text{ }^{\circ}\text{C}$. Which statements below are correct? Neglect the heat capacity of the cup and assume that there is no heat exchange with the surroundings.

- (1) The average kinetic energy of the water molecules in the tea decreases.
- (2) The average potential energy of the water molecules in the tea remains unchanged.
- (3) The energy lost by the tea equals the energy gained by the milk.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

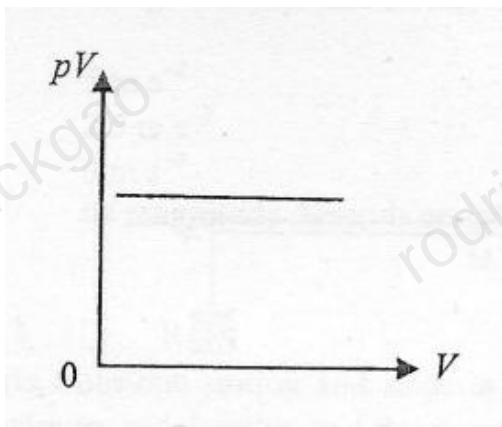
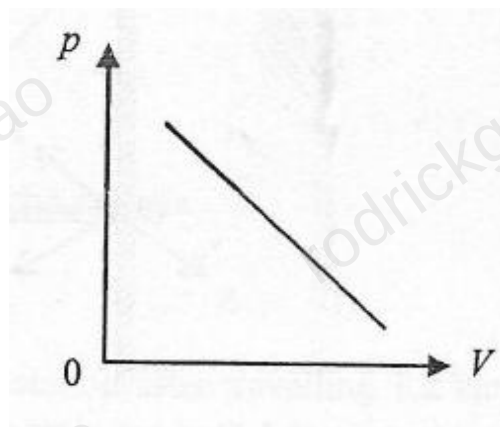
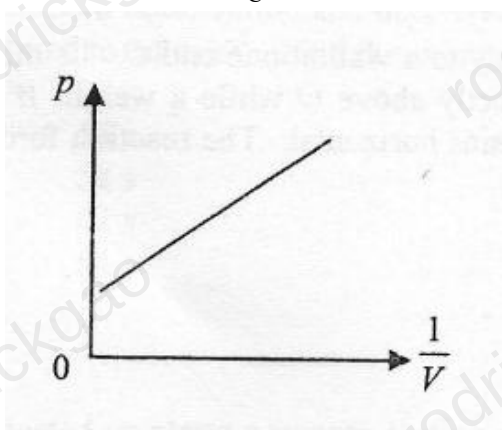
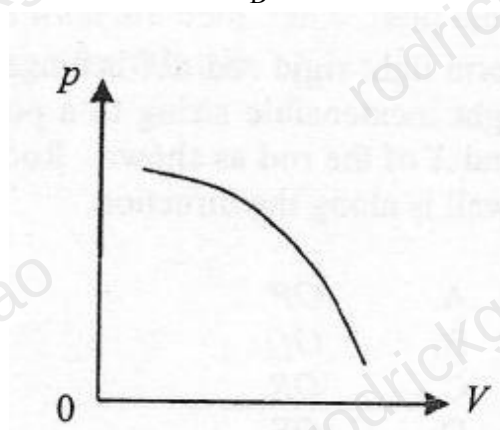
D (1), (2) and (3)

Official answer: D

The tea cools, so the average molecular kinetic energy decreases. With no change of state, the average intermolecular potential energy is taken as unchanged; by energy conservation, energy lost by the tea equals energy gained by the milk. Answer: D.

Question 4 *

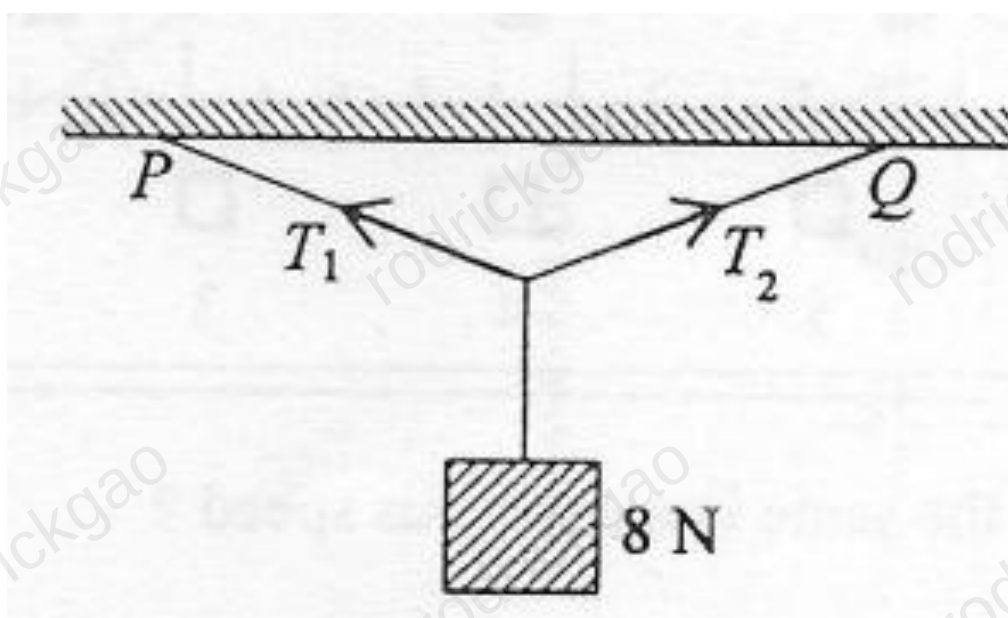
From which graph below can one deduce that the pressure p of a fixed mass of an ideal gas is inversely proportional to its volume V when the temperature of the gas is kept constant?

A**B****C****D****Official answer: A**

For a fixed mass of gas at constant temperature, $pV = \text{constant}$, so $p \propto 1/V$. A horizontal pV - V graph therefore shows that pV remains constant. Answer: A.

Question 5

A block of weight 8 N is suspended from a horizontal ceiling by light inextensible strings to two different points P and Q as shown. The strings are equal in length. Which of the following descriptions about the tensions T_1 and T_2 in the two strings is/are correct?



- (1) The magnitude of T_1 must be greater than 4 N.
- (2) The maximum value of T_2 would not exceed 8 N.
- (3) The resultant force of T_1 and T_2 is zero.

A (1) only

B (3) only

C (1) and (2) only

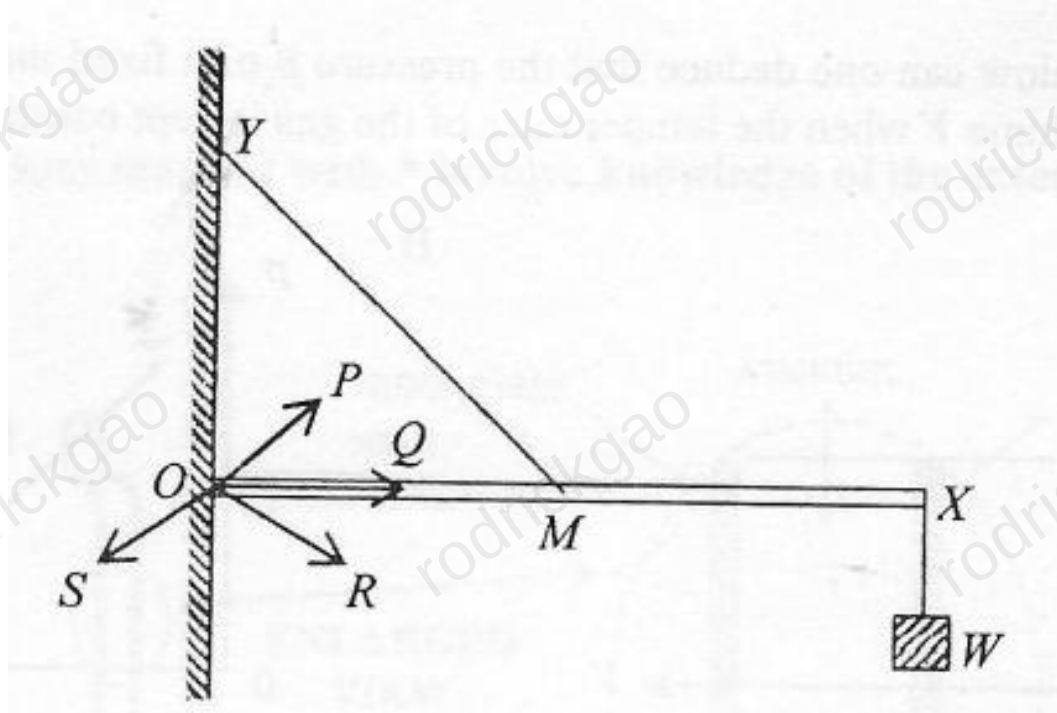
D (2) and (3) only

Official answer: A

Vertical equilibrium requires the upward components of T_1 and T_2 to total 8 N, so T_1 must exceed its vertical component and hence exceed 4 N. T_2 can exceed 8 N for a sufficiently shallow string, and $T_1 + T_2$ balances the weight rather than summing to zero. Answer: A.

Question 6

A uniform light rigid rod OX is hinged smoothly to a wall at one end O . Its mid-point M is connected by a light inextensible string to a point Y directly above O while a weight W is suspended from the other end X of the rod as shown. Rod OX remains horizontal. The reaction force acting on the rod due to the wall is along the direction



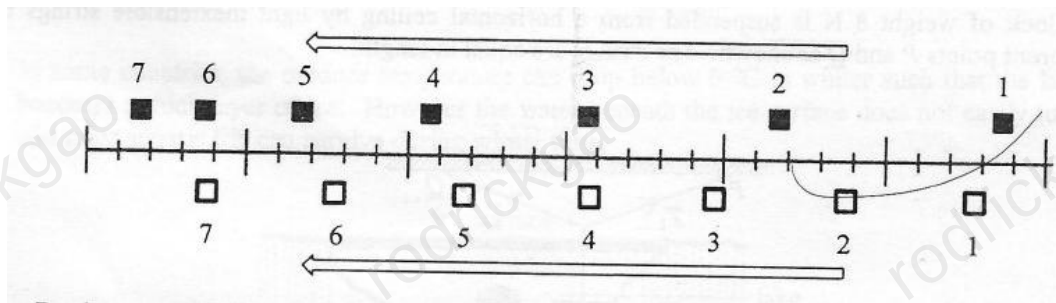
- A OP
- B OQ
- C OR**
- D OS

Official answer: C

Taking moments about O , the vertical component of the string tension at M satisfies $T_y(L/2) = WL$, hence $T_y = 2W$. Force balance then requires the hinge reaction to act downwards and to the right, along OR . Answer: C.

Question 7

Two blocks X and Y are moving towards the left. Their positions at successive instants (indicated by the numbers) of equal time intervals are shown below. Do these two blocks ever have the same instantaneous speed?



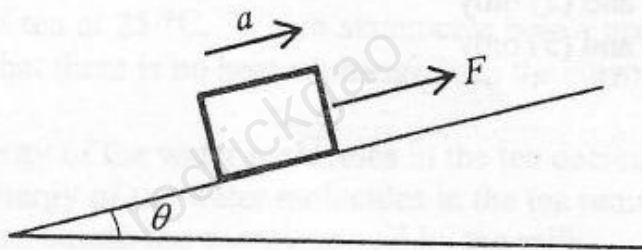
- A Yes, at instant 3.
- B Yes, at a moment between instants 4 and 5.
- C Yes, at instant 6.
- D No.

Official answer: B

The spacing of X's successive positions decreases while that of Y changes in the opposite sense. Their interval-average speeds pass from one ordering to the other, so by continuity their instantaneous speeds are equal at some time between instants 4 and 5. Answer: B.

Question 8

A block of mass m is placed on a smooth incline making an angle θ with the horizontal as shown. When a force of magnitude F parallel to the incline is applied to the block, it travels up the incline with an acceleration a . If the applied force becomes $2F$, what would the magnitude of the acceleration be?



- A greater than $2a$
- B equal to $2a$
- C between a and $2a$
- D whether it is greater than $2a$, equal to $2a$ or between a and $2a$ depends on the value of θ

Official answer: A

Initially $F - mg \sin \theta = ma$, so $F/m = a + g \sin \theta$. With force $2F$, $a' = 2F/m - g \sin \theta = 2a + g \sin \theta$, which is greater than $2a$. Answer: A.

Question 9

A particle moving along a straight line with speed 0.5 m s^{-1} changes its direction of motion within a period of 0.2 s and subsequently travels with the same speed in the opposite direction. Find the magnitude of its average acceleration during this 0.2 s .

- A 5 m s^{-2}
- B 2.5 m s^{-2}
- C 0 m s^{-2}
- D its magnitude depends on the mass of the particle

Official answer: A

Taking right as positive, $\Delta v = -0.5 - (+0.5) = -1.0 \text{ m s}^{-1}$. Thus $|\bar{a}| = |\Delta v|/\Delta t = 1.0/0.2 = 5 \text{ m s}^{-2}$. Answer: A.

Question 10

A train departs from one station and stops at the next station after travelling 1.2 km along a straight line. The maximum acceleration and deceleration of the train are both 5 m s^{-2} and the highest speed of the train is limited to 20 m s^{-1} . Find the shortest time taken for this trip.

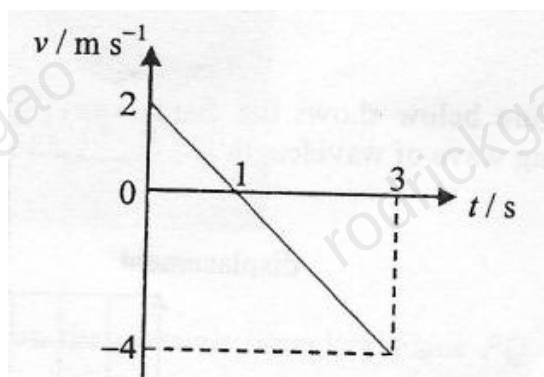
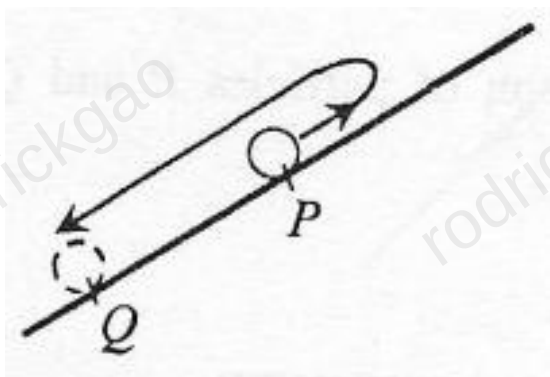
- A 56 s
- B 58 s
- C 62 s
- D 64 s

Official answer: D

The train accelerates to 20 m s^{-1} in 4 s and covers 40 m; it needs the same time and distance to stop. It cruises through the remaining $1200 - 80 = 1120 \text{ m}$ in 56 s, so the minimum time is $4 + 56 + 4 = 64 \text{ s}$. Answer: D.

Question 11

A bead is projected up along a smooth incline at point P at time $t = 0$. After reaching the highest point, the bead then travels downwards and passes point Q at $t = 3 \text{ s}$ as shown. The velocity–time (v – t) relationship of the bead is shown in the graph below. Find the separation PQ along the incline.



- A 2 m
- B 3 m
- C 4 m
- D 5 m

Official answer: B

The displacement from P to Q is the signed area under the v – t graph from 0 to 3 s. The positive and negative triangular areas give a net displacement of 3 m in magnitude. Answer: B.

Question 12

At a certain instant, an object flying horizontally to the right at 1 m s^{-1} suddenly explodes into two fragments of mass ratio 1: 2. If just after the explosion the more massive fragment flies to the right at 3 m s^{-1} , what would happen to the other fragment just after the explosion?

- A It would fly to the left at 3 m s^{-1} .
- B It would fly to the left at 4 m s^{-1} .
- C It would be momentarily at rest.
- D It would fly to the right at 1 m s^{-1} .

Official answer: A

Let the fragment masses be m and $2m$. Momentum conservation gives $3m(1) = m v + 2m(3)$, so $v = -3 \text{ m s}^{-1}$; the lighter fragment moves left. Answer: A.

Question 13 *

A satellite of mass m moves around a planet of mass M in a circular orbit of radius r . What does the angular velocity of the satellite depend on?

- (1) r
- (2) m
- (3) M

- A (1) only
- B (2) only
- C (1) and (3) only
- D (2) and (3) only

Official answer: C

For a circular orbit, $GMm/r^2 = m\omega^2 r$, hence $\omega = \sqrt{GM/r^3}$. It depends on M and r , but not on the satellite mass m . Answer: C.

Question 14

Which of the following statements about waves is/are correct?

- (1) Longitudinal waves can transmit energy from one place to another but transverse waves cannot.
- (2) Sound waves propagate faster in water than in air.
- (3) Infra-red radiation is a kind of electromagnetic wave.

A (1) only

B (3) only

C (1) and (2) only

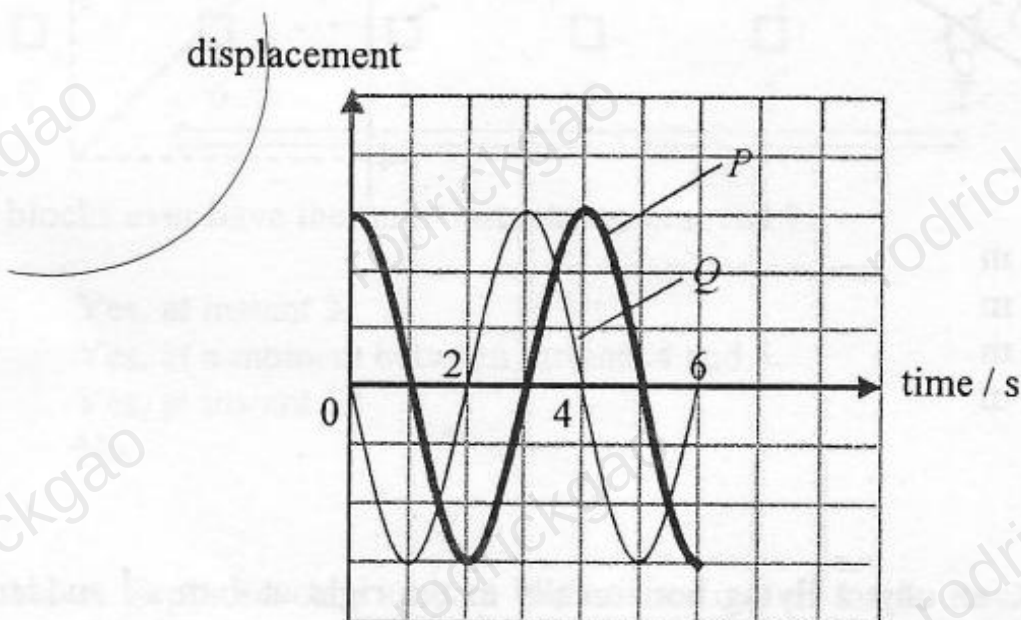
D (2) and (3) only

Official answer: D

Both longitudinal and transverse waves transfer energy, so (1) is false. Sound travels faster in water than in air, and infrared radiation is electromagnetic; (2) and (3) are correct. Answer: D.

Question 15

The figure below shows the displacement–time graph of particles P and Q on the same transverse travelling wave of wavelength λ . Which of the following statements MUST BE correct? Upward displacement is taken to be positive.



- (1) At time $t = 2$ s, P is momentarily at rest.
- (2) At time $t = 4$ s, Q is moving downwards.
- (3) The separation between the equilibrium positions of P and Q is 0.25λ .

A (2) only

B (3) only

C (1) and (2) only

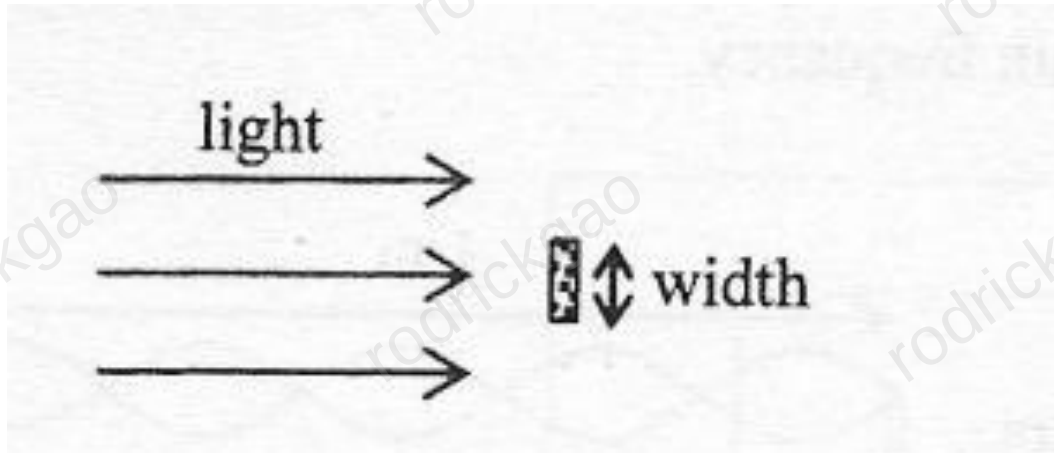
D (1) and (3) only

Official answer: C

At $t = 2$ s, P is at an extreme displacement and is momentarily at rest. At $t = 4$ s the Q curve has negative slope, so Q is moving downwards. Phase information alone does not fix the separation as exactly 0.25λ . Answer: C.

Question 16

Light undergoes diffraction round an obstacle. The angle of diffraction would increase when



- (1) the amplitude of the incident light is increased.
- (2) the width of the obstacle is increased.
- (3) the wavelength of the incident light is increased.

A (1) only

B (3) only

C (1) and (2) only

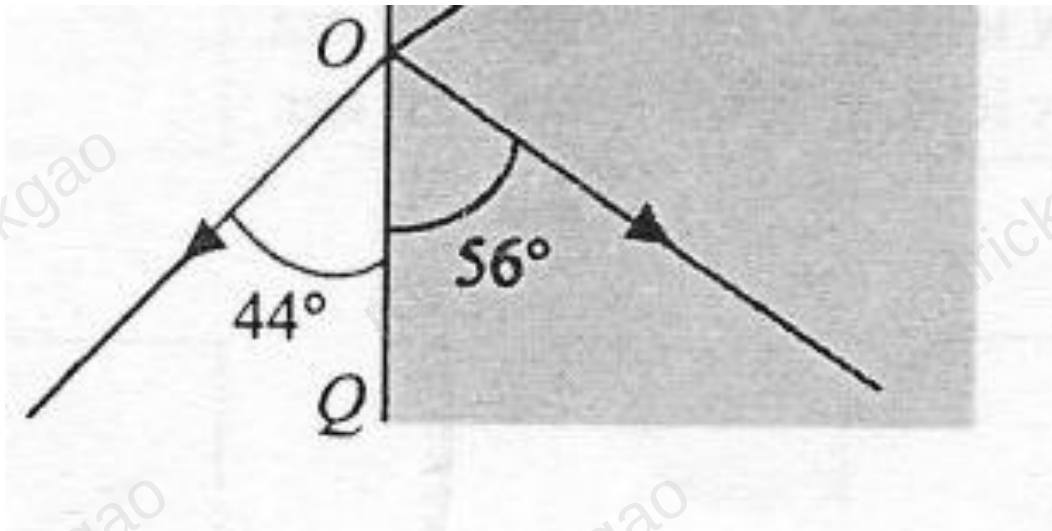
D (2) and (3) only

Official answer: B

Diffraction becomes more pronounced when wavelength increases relative to obstacle size. Amplitude does not set the diffraction angle, and increasing the obstacle width reduces it; only (3) is correct. Answer: B.

Question 17

In the figure above, XO is a light ray incident on the glass–air boundary plane PQ. Which of the following gives the refractive index of glass?



- A $\sin 56^\circ / \sin 44^\circ$
- B $\sin 44^\circ / \sin 34^\circ$
- C $\sin 56^\circ / \sin 46^\circ$
- D $\sin 46^\circ / \sin 34^\circ$

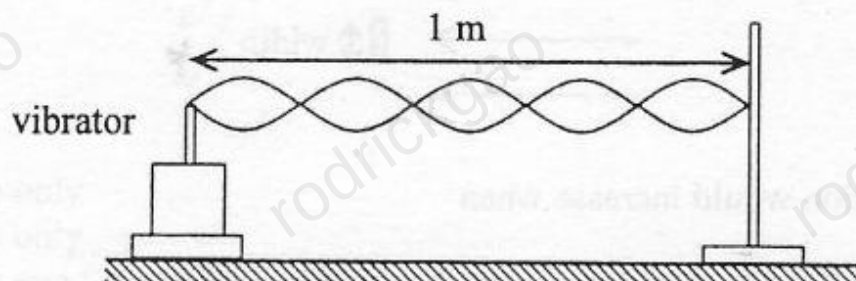
Official answer: D

The angles to the normal are $i = 34^\circ$ in glass and $r = 46^\circ$ in air. Snell's law gives $n = \sin r / \sin i = \sin 46^\circ / \sin 34^\circ$.

Answer: D.

Question 18

The figure shows a string with one end fixed and the other end tied to a vibrator. A stationary wave is formed as shown at a certain frequency. If the speed of the wave along the string is 7 m s^{-1} , what is the frequency of the wave?



- A 2.8 Hz
- B 7 Hz
- C 17.5 Hz
- D 35 Hz

Official answer: C

From the stationary-wave pattern, adjacent nodes are 0.20 m apart, so $\lambda/2 = 0.20 \text{ m}$ and $\lambda = 0.40 \text{ m}$. Therefore $f = v/\lambda = 7/0.40 = 17.5 \text{ Hz}$. Answer: C.

Question 19 *

An object placed 25.0 cm in front of a lens forms a virtual image at a distance 11.1 cm from the lens. The lens is a

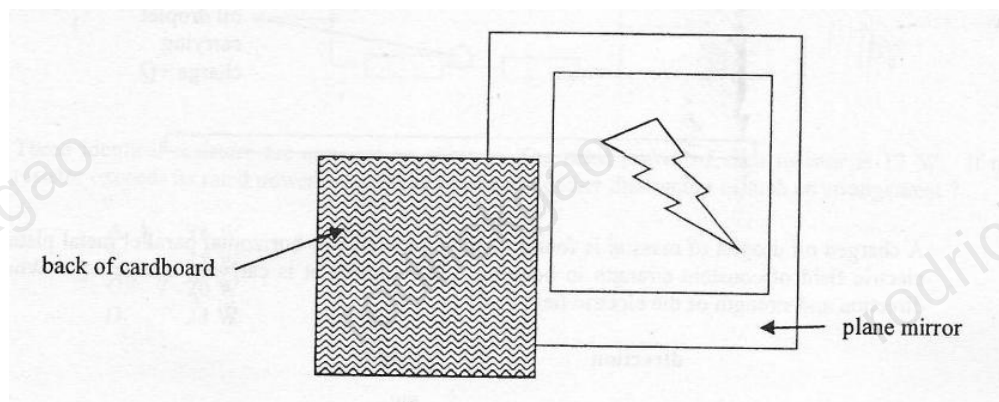
- A concave lens of focal length 7.7 cm.
- B concave lens of focal length 20 cm.
- C convex lens of focal length 7.7 cm.
- D convex lens of focal length 20 cm.

Official answer: B

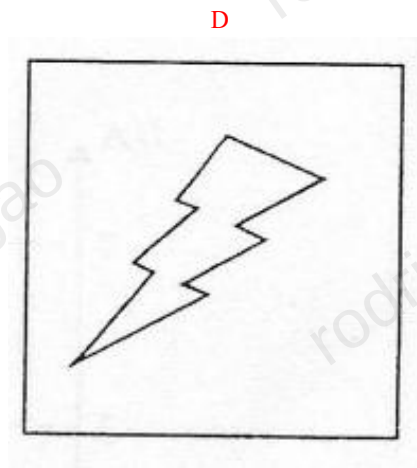
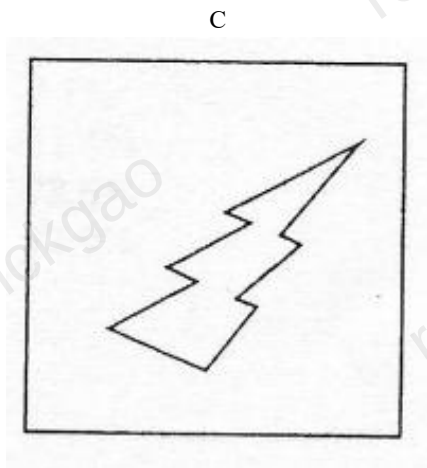
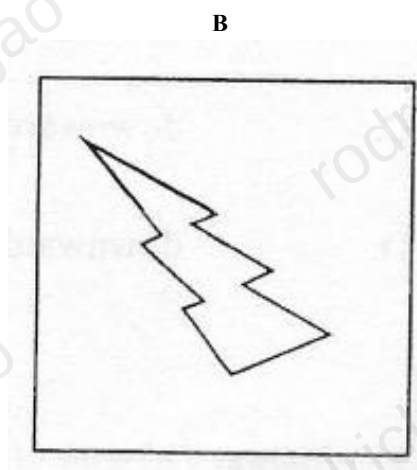
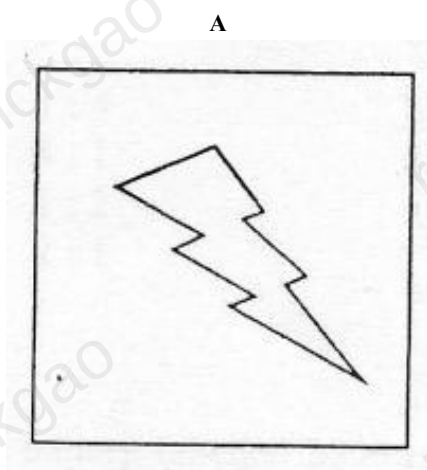
Using the lens sign convention, $u = 25.0 \text{ cm}$ and $v = -11.1 \text{ cm}$ for a virtual image. Then $1/f = 1/u + 1/v$, giving $f \approx -20 \text{ cm}$; the negative focal length identifies a concave lens. Answer: B.

Question 20

The figure shows the image seen when a plane mirror is placed in front of a cardboard with a design on its front surface. Which diagram below shows the design on the cardboard?



Mirror and cardboard arrangement



Official answer: D

A plane mirror reverses front-to-back while preserving distances and orientation parallel to the mirror. Reversing the shown mirror image reconstructs the design in option D. Answer: D.

Question 21

Which of the following is NOT a typical sound intensity level that occurs in daily life?

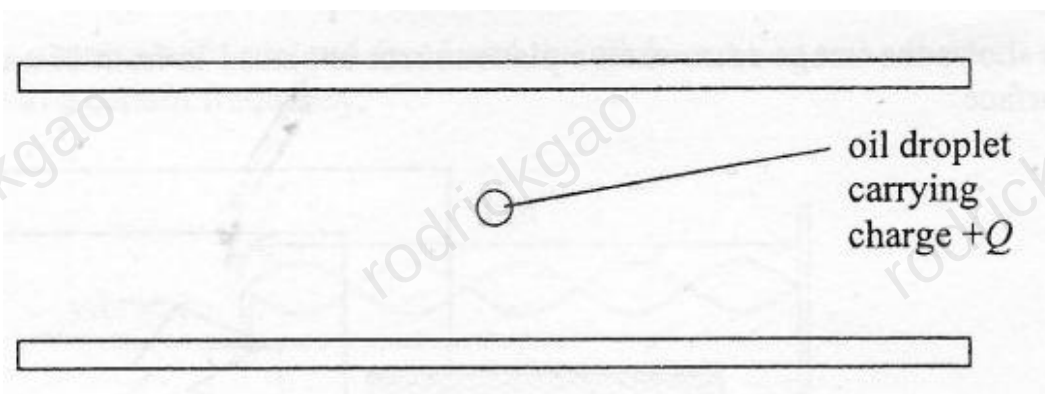
- A 130 dB: when an airplane takes off
- B 110 dB: at a rock concert
- C 80 dB: having a normal conversation
- D 30 dB: inside a library

Official answer: C

A normal conversation is typically about 60 dB, not 80 dB. The other quoted levels are reasonable daily-life values. Answer: C.

Question 22

A charged oil droplet of mass m is found floating between two horizontal parallel metal plates with an electric field of constant strength in between. The oil droplet is carrying charge $+Q$. What are the direction and strength of the electric field?



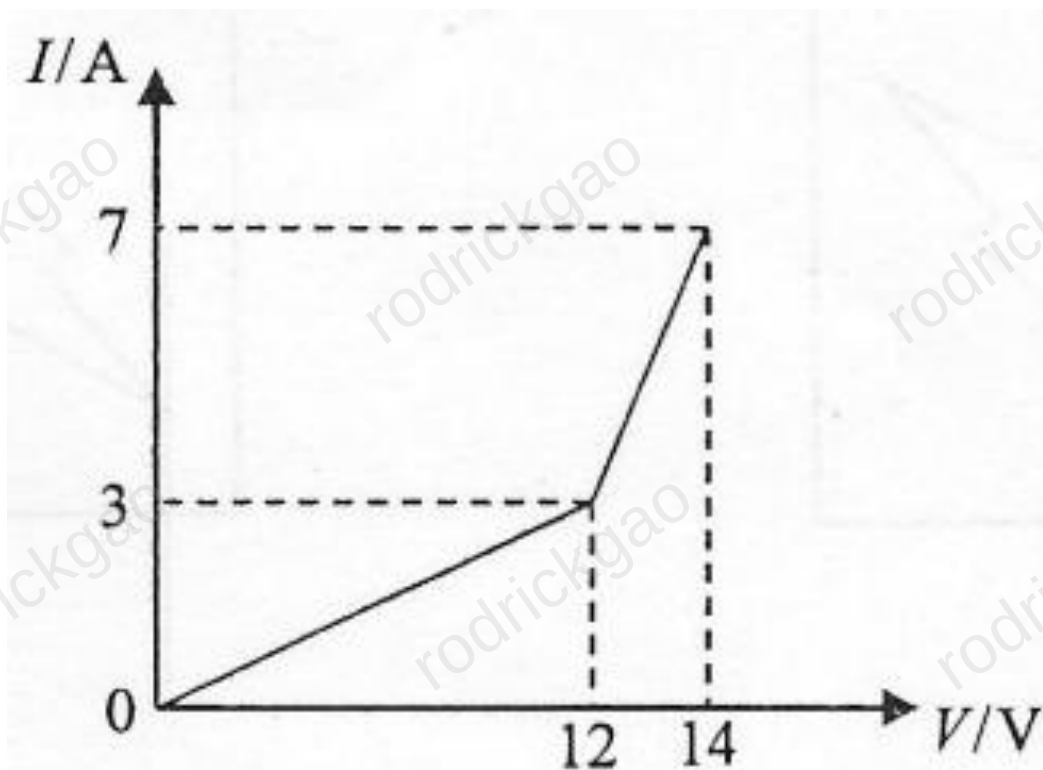
	Direction	Strength
A	upward	mg/Q
B	upward	$Q/(mg)$
C	downward	mg/Q
D	downward	$Q/(mg)$

Official answer: A

For the positive droplet to float, the electric force QE must be upward and equal to mg . Hence $E = mg/Q$ and the field is upward. Answer: A.

Question 23

The graph below shows the current–voltage (I–V) relationship of a conductor. Which statement below is INCORRECT?



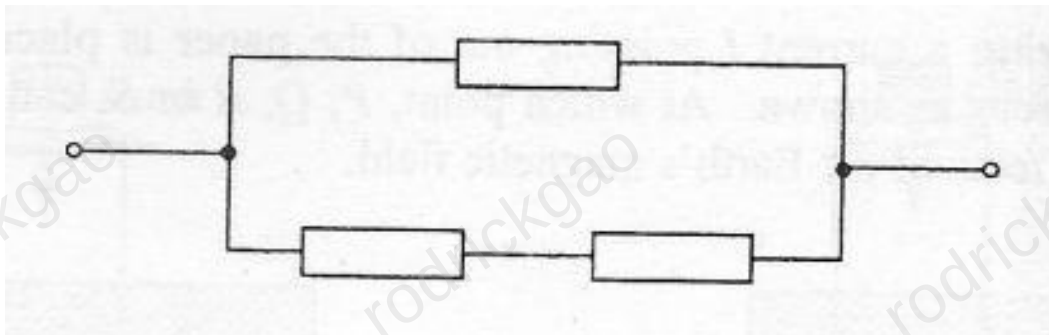
- A When the voltage across the conductor is less than 12 V, it obeys Ohm's law.
- B When the voltage across the conductor exceeds 12 V, its resistance begins to decrease.
- C The resistance of the conductor is $0.5\ \Omega$ when the current flowing through it is 5 A.
- D The resistance of the conductor is $2\ \Omega$ when the voltage across it is 14 V.

Official answer: C

At $I = 5\text{ A}$, the graph gives a voltage of about 10 V, so $R = V/I \approx 2\ \Omega$, not $0.5\ \Omega$. Thus statement C is the incorrect one. Answer: C.

Question 24

Three identical resistors are arranged as shown. The rated power of each resistor is 12 W. If no resistor exceeds its rated power, what is the maximum power dissipation in such an arrangement?



- A 16 W
- B 18 W**
- C 20 W
- D 24 W

Official answer: B

At the limiting voltage, the single resistor in the upper branch dissipates its rated 12 W. Each identical resistor in the lower series branch has half the voltage and dissipates $12/4 = 3$ W, so the total is $12 + 3 + 3 = 18$ W.

Answer: B.

Question 25

Two wires X and Y of the same length are made from different materials. The radius of X is half that of Y. When the two wires are connected in parallel to the same power supply, the current flowing through each wire is the same. What is the ratio of the resistivity of the material of X to that of Y?

- A 1: 4**
- B 4: 1
- C 1: 2
- D 2: 1

Official answer: A

Equal currents in parallel under the same voltage mean $R_x = R_y$. Since $R = \rho L/A$ and $A_x = A_y/4$, equality requires $\rho_x = \rho_y/4$. Thus $\rho_x:\rho_y = 1:4$. Answer: A.

Question 26

A mobile phone battery labelled 2800 mA h capacity is fully charged initially. What percentage of capacity remains after it has delivered a current of 200 mA for 3 hours?

- A 7.1%
- B 21.4%
- C 78.6%**
- D 92.9%

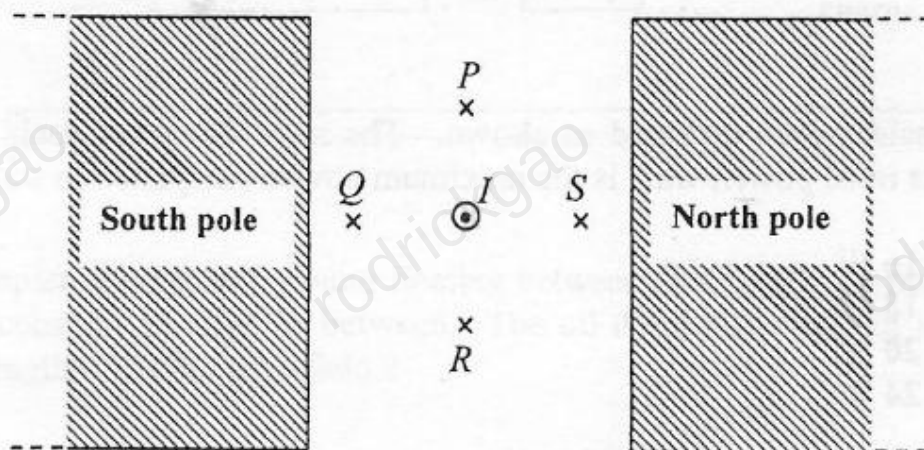
Official answer: C

Charge used = $200 \text{ mA} \times 3 \text{ h} = 600 \text{ mA h}$. The remaining fraction is $(2800 - 600)/2800 = 0.7857$, i.e. 78.6%.

Answer: C.

Question 27

A straight wire carrying a current I pointing out of the paper is placed in a uniform magnetic field between two pole pieces as shown. At which point, P, Q, R or S, can the resultant magnetic field be zero? Neglect the effects of the Earth's magnetic field.



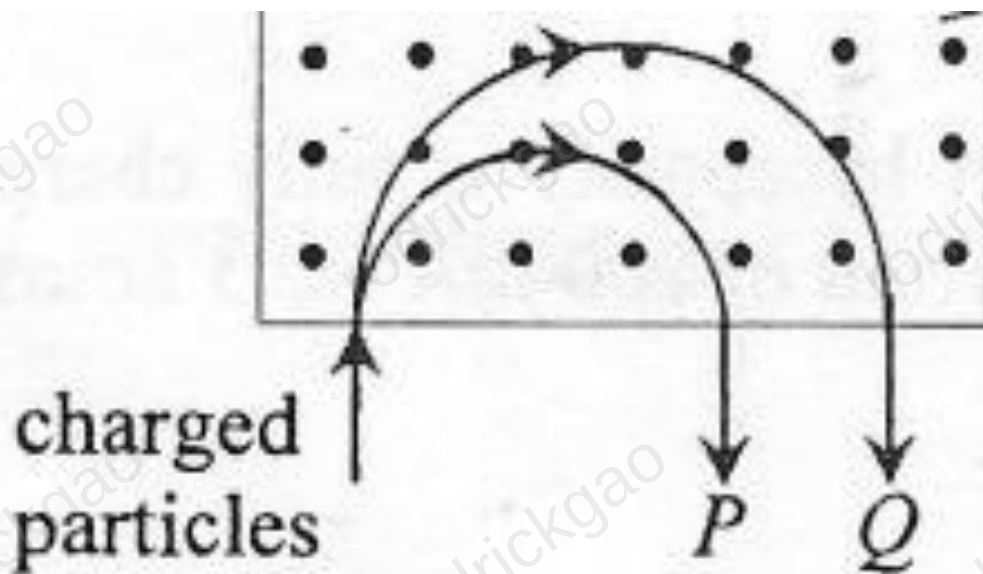
- A P
- B Q
- C R**
- D S

Official answer: C

The uniform field points from the north pole to the south pole, i.e. leftwards. For current out of the page, the wire's field is anticlockwise and points rightwards below the wire, so cancellation can occur at R. Answer: C.

Question 28 *

P and Q are two particles carrying the same amount of charge but of different masses. They travel with the same speed and enter a uniform magnetic field pointing out of the paper as shown. Semi-circular paths with different radii are described before they emerge from the field. Which descriptions below are correct?



- (1) Both P and Q are positively charged.
- (2) P and Q emerge from the field with the same speed.
- (3) The mass of Q is greater than that of P.

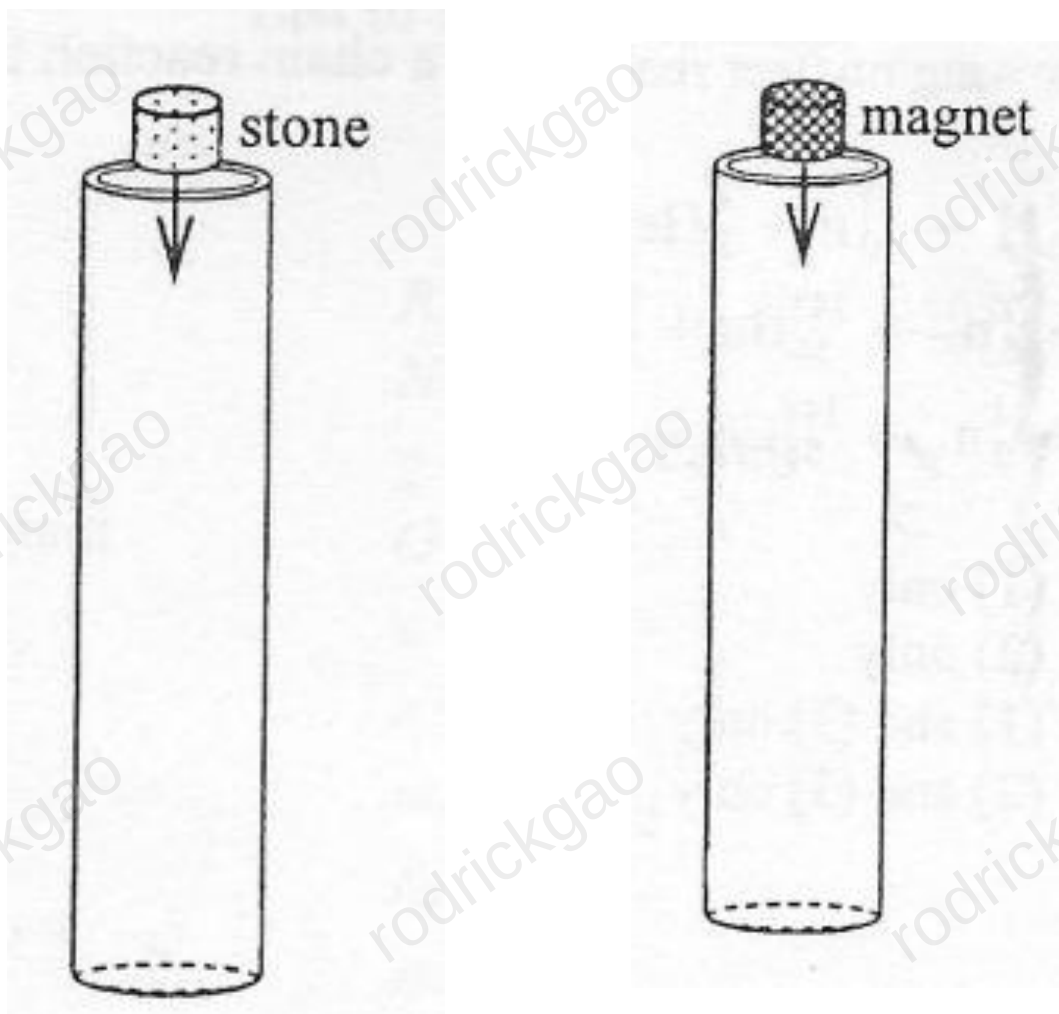
- A (1) and (2) only
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

Official answer: D

The paths bend to the right for upward motion in an outward field, so both charges are positive. Magnetic force does no work, so both emerge at their original common speed; with $r = mv/(qB)$, Q's larger radius means greater mass. Answer: D.

Question 29

A stone and a strong magnet of the same size and shape are released from rest into a hollow aluminium tubing respectively. Which of the following is correct? Neglect air resistance.



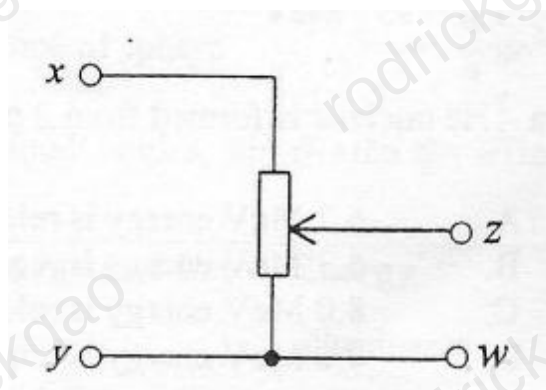
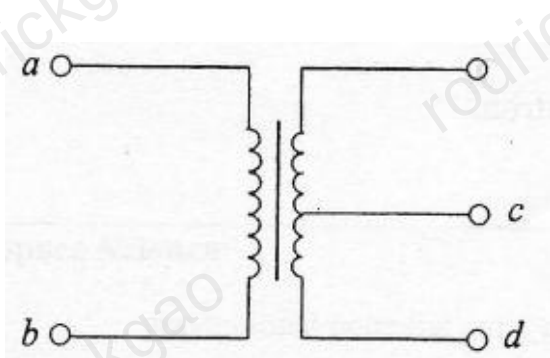
	Drops more slowly	Reason
A	stone	the stone is more massive
B	magnet	the stone is more massive
C	stone	the magnet induces eddy current in the aluminium tubing
D	magnet	the magnet induces eddy current in the aluminium tubing

Official answer: D

The falling magnet changes the magnetic flux through the aluminium and induces eddy currents. By Lenz's law their magnetic effect opposes the fall, so the magnet drops more slowly. Answer: D.

Question 30 *

In the circuits below, if a 12 V sinusoidal a.c. is applied across ab and across xy respectively, the voltages across cd and zw are both 6 V. Now if a 6 V sinusoidal a.c. is applied across cd and across zw respectively, what would be the voltages across ab and xy respectively?



	Voltage across ab	Voltage across xy
A	12 V	12 V
B	12 V	6 V
C	6 V	6 V
D	12 V	0 V

Official answer: B

A transformer is reciprocal: 6 V applied to winding cd gives 12 V across ab. For the potentiometer, with no load at x the section xz carries no current, so x is at the same potential as z and 6 V appears across xy. Answer: B.

Question 31

Which of the following nuclear reactions is/are possible for a chain reaction to occur?

- (1) ${}^2_1\text{H} + {}^3_1\text{H} \rightarrow {}^1_0\text{n} + {}^4_2\text{He}$
- (2) ${}^{235}_{92}\text{U} + {}^1_0\text{n} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3 {}^1_0\text{n}$
- (3) ${}^{239}_{94}\text{Pu} + {}^1_0\text{n} \rightarrow {}^{148}_{58}\text{Ce} + {}^{89}_{36}\text{Kr} + 3 {}^1_0\text{n}$

- A (1) only
- B (2) only
- C (1) and (3) only
- D (2) and (3) only

Official answer: D

A chain reaction requires each event to release neutrons that can trigger further events. The two fission reactions (2) and (3) each release three neutrons, whereas the fusion reaction (1) does not provide such multiplication.

Answer: D.

Question 32 *

X and Y are two radioactive nuclides. The ratio of the mass of an atom of X to that of an atom of Y is 1: 2. The half-lives of X and Y are T and 2T respectively. If two samples consisting of purely X and Y respectively have the same initial mass, find the ratio of the number of undecayed nuclei of X to that of Y after a period of 4T.

- A 1: 4
- B 1: 2
- C 1: 1
- D 2: 1

Official answer: B

For equal initial sample masses and atomic-mass ratio 1:2, $N_{x0} = 2N_{y0}$. After 4T, $N_x = 2N_{y0}(1/2)^4 = N_{y0}/8$, while $N_y = N_{y0}(1/2)^2 = N_{y0}/4$; hence $N_x:N_y = 1:2$. Answer: B.

Question 33 *

Given: mass of proton = 1.007276 u; mass of neutron = 1.008665 u; mass of ${}^3_2\text{He}$ nucleus = 3.016030 u; 1 u = 931 MeV. When a ${}^3_2\text{He}$ nucleus is formed from 2 protons and 1 neutron,

- A 6.7 MeV energy is released.
- B 6.7 MeV energy is required.
- C 8.0 MeV energy is released.
- D 8.0 MeV energy is required.

Official answer: A

Mass defect $\Delta m = 2(1.007276) + 1.008665 - 3.016030 = 0.007187$ u. The binding energy released is $\Delta mc^2 = 0.007187 \times 931 \approx 6.7$ MeV. Answer: A.